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Assignment 1

Question 1:

1.1

- a) In this scenario the hidden-technical-debt that best explains the mistake committed by the development team is Entanglement (Changing Anything Changes Everything). From a software engineering perspective, a developer should generally aim at writing low coupled and highly cohesive code, that is, each function/file should have a specific purpose that works as a “unit” and interacts with others cooperatively for the bigger goal of the code but without depending on them to perform their own specific objective. When this does not happen, even in the slightly different context that is ML code, one specific section of code, such as the one mentioned in this example, is subject to unintended changes due to an alteration elsewhere in the project.
- b) Undeclared consumers. Due to the unnotified use of the output of the models output by the marketing team, the use itself becomes very inflexible, that is, the use of the raw data from the table would not be of the same use to the marketing team if the table that contains this data undergoes changes made by the ML team for whatever reason they might need to alter the models output table.
- c) Configuration & Glue-Code Debt. When we have an environment such as the one described in the item, error becomes highly prone to happen eventually, if not by the team that is currently managing the MLops, then by future teams. That is, if there is no kind of coordination and general guidelines regarding how to maintain and reproduce the results in an orderly fashion, it becomes very hard for teams to properly use model.

1.2

Regarding the third scenario of item 1.1, one specific tool that could be used to better orchestrate the ML pipeline is MLflow. Once implemented, the messy scripts would be substituted by proper description of the environment inside the MLproject YAML file.

